A symmetry has two meanings for a mixed state

Strong symmetry

$U \rho = e^{i\alpha} \rho$
One global charge sector
Every ensemble vector shares charge

implies

Weak symmetry

$U \rho U^* = \rho$ (* = adjoint)
The ensemble is invariant
Different charge sectors may mix

Strong-to-weak spontaneous order

Exact strong symmetry at finite size; long-range nonlinear overlap.
Ordinary charged two-point correlations decay.

Spontaneous order does not mean the finite-size density matrix loses its exact symmetry.